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# PHYSICS PTA - MODEL QUESTION PAPER -2 – QUESTIONS AND ANSWERS
## STANDARD XII
## PART – I

**15× 1 =15**

1. Hydrogen atom in ground state is excited by a monochromatic radiation of wavelength of $\lambda = 975$ Å. Number of spectral lines in the resulting spectrum emitted will be:

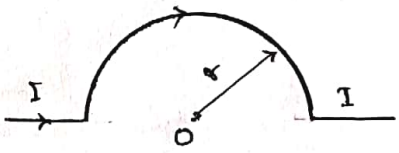
a) 10 **b) 6** c) 3 d) 2

**Hints:** $E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{975\times10^{-10}\times1.6\times10^{-19}}$ =12.75 eV ; $E_n$ = -(13.6 + 12.75) eV = - 0.85 eV

$E_n = \frac{-13.6\ eV}{n^2}$ ; $n^2 = \frac{-13.6}{-0.85} = 16$ ; n = 4; Number of spectral line = $\frac{n(n-1)}{2} = \frac{4\times3}{2} = 6$

2. The magnetic field at the centre O of the following current loop is

a) $\frac{\mu_0 I}{2r}\otimes$ b) $\frac{\mu_0 I}{4r}\odot$ c) $\frac{\mu_0 I}{2r}\odot$ **d)** $\frac{\mu_0 I}{4r}\otimes$




3. Two photons, each of energy 2.5eV are simultaneously incident on sodium metal surface. If the work function of sodium is 2.75eV, then from its surface ___ will be emitted.

a) one electron b) two electrons
c) more than two electrons **d) not a single electron**

4. Stars twinkle due to:

a) total internal reflection b) reflection of light
**c) refraction of light** d) diffraction

5. A carbon resistor of (47± 4.7) kΩ to be marked with rings of different colours for its identification. The colour code sequence will be :

a) Yellow – Green – Violet – Gold **b) Yellow – Violet – Orange – Silver**
c) Violet – Yellow - Orange – Silver d) Green – Orange – Violet – Gold

6. An electric generator consists of 10-turn square wire loop of side 50cm. The loop is turned so as to produce 50Hz A.C. How strong must be magnetic field be for the peak output voltage to be 300V?

a) 2.4T b) 0.417T c) 2.62 T **d) 0.382 T**

**Hints :** $\varepsilon_m = NBA\omega = NBA\,(2\pi f)$ ; $B = \frac{\varepsilon_m}{NA.2\pi f} = \frac{300}{10\times(0.5)^2\times2\times\pi\times50} = 0.382$ T

7. The unit of permittivity is:

**a) $C^2N^{-1}m^{-2}$** b) $N\,m^2C^{-2}$ c) $H\,m^{-1}$ d) $N\,C^{-2}m^{-2}$

8. The signal is affected by noise in a communication system:

a) at the modulator **b) in the channel** c) at the receiver d) at the transmitter

9. A photoelectric surface is illuminated successively by monochromatic light of wavelength $\lambda$ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times than in the first case, the work function at the surface of material is:

a) $\frac{hc}{\lambda}$ b) $\frac{2hc}{\lambda}$ c) $\frac{hc}{3\lambda}$ **d)** $\frac{hc}{2\lambda}$

**Hints:** $\frac{hc}{\lambda} = W + KE$ .....(1) $\frac{hc}{(\frac{\lambda}{2})} = W + 3KE$ .....(2)

(1) × 3 $\frac{3hc}{\lambda} = 3W + 3KE$.....(3) ; (3) – (2) $\frac{hc}{\lambda} = 2W$; $W = \frac{hc}{2\lambda}$

10. Which one of the following is a natural nanomaterial?

**a) Peacock feather** b) Peacock beak c) Grain of sand d) Skin of the whale


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11. In Joule's heating law, when R and t are constant, if the H is taken along the y-axis and I along the x-axis, the graph is:
**a) straight line** b) parabola c) circle d) ellipse

12. The specific characteristic of a common emitter amplifier is :
**a) signal phase reversal** b) low current gain
c) high input resistance d) low power gain

13. A coil of area of cross-section 0.5 $m^2$ with 10 turns is in a plane which is parallel to a uniform electric field of 100N/C. The flux through the plane is
a) 100 V.m b) 500 V.m c) 20 V.m **d) zero**
**Hints :** $\phi = NBA\cos\theta$; $\theta = 90°$, $\cos 90 = 0$, $\phi = 0$

14. Electromagnetic waves are
a) mechanical waves only b) longitudinal waves only
c) both mechanical and longitudinal **d) transverse but non- mechanical**

15. When the number of turns in a galvanometer is doubled,
a) both current sensitivity and voltage sensitivity are doubled
b) current sensitivity does not change but voltage sensitivity is doubled
**c) voltage sensitivity does not change but current sensitivity is doubled**
d) both current sensitivity and voltage sensitivity do not change

## PART – II

**Answer any six questions . Question number 24 compulsory.** $6 \times 2 = 12$

**16. The refractive index of a prism material is 1.541. Find its critical value.**

$\sin i_c = \frac{1}{n}$; $\sin i_c = \frac{1}{1.541} = 0.6489$; $i_c = \sin^{-1}\left(\frac{1}{n}\right) = \sin^{-1}(0.6489) = 42°27'$

**17. How a Galvanometer can be converted in to an ammeter and a Volt meter?**
A galvanometer is converted into an ammeter by connecting a low resistance in parallel with the galvanometer.
A galvanometer is converted into a voltmeter by connecting high resistance $R_h$ in series with galvanometer

**18. What is the value of charge of a neutrino? Although trillions of neutrons coming from the sun pass through our body every second, it is very difficult to detect them. Why?**
Neutrino has no charge.
Recent experiments showed that the neutrino has very tiny mass.
It interacts very weakly with the matter. Therefore, it is very difficult to detect. In fact, in every second, trillions of neutrinos coming from the sun are passing through our body without any interaction

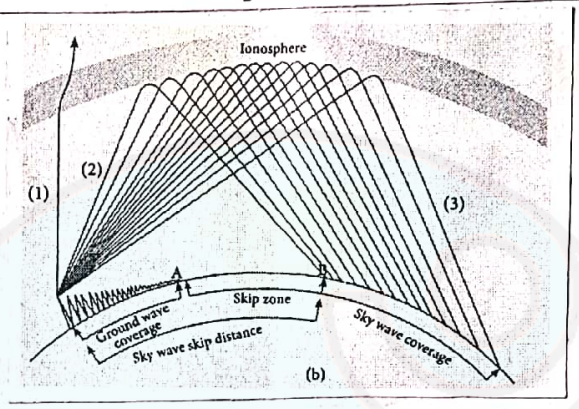
**19. Write a short note on absorption spectrum.**
When light is allowed to pass through a medium or an absorbing substance then the spectrum obtained is known as absorption spectrum. It is the characteristic of absorbing substance.
Absorption spectrum is classified into three types:
(i) Continuous absorption spectrum (ii) Line absorption spectrum (iii)band absorption spectrum

**20. Calculate the momentum of the electron with kinetic energy 2 eV**

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$$p = \sqrt{2mK} = \sqrt{2 \times 9.11 \times 10^{-31} \times 2 \times 1.6 \times 10^{-19}}$$
$$= 2 \times 10^{-25} \times \sqrt{14.576}$$
$$= 2 \times 3.818 \times 10^{-25}$$
$$= 7.636 \times 10^{-25} \text{ kg m s}^{-1}$$

**21. Show in diagram, skip distance and skip zone.**




**22. State the law of conservation of electric charges**

The total electric charge in the universe is constant and charge can neither be created nor be destroyed. In any physical process, the net change in charge will always be zero.

**23. State De Morgan's theorems**

**De Morgan's First Theorem**

The first theorem states that the complement of the sum of two logical inputs is equal to the product of its complements. $\overline{A+B} = \overline{A}.\overline{B}$

**De Morgan's Second Theorem**

The second theorem states that the complement of the product of two inputs is equal to the sum of its complements. $\overline{A.B} = \overline{A} + \overline{B}$

**24. When an inductor is connected to a 230 V d.c. source of 2A passes through it. When the same inductor is connected to a 230V, 50 Hz a.c. source , the amount of current decreases (i.e, 1 A)Why?**

The formula for calculating impedance in case of a conductor having inductance L is

$$X_L = 2\pi f. L$$

Now in the case of DC source , f will be equal to zero. Hence $X_L = 0$, when the impedance or resistance is less, more current 2A passes through the circuit.

But in the case of AC source of f = 50 Hz, then there will be some impedance or resistance added to the circuit. It reduces the overall current to 1A

## PART - III

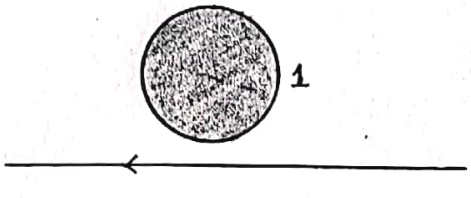
**Answer any six questions . Question number 33 compulsory.** **6 × 3 = 18**

**25. i) State Lenz's law.**

Lenz's law states that the direction of the induced current is such that it always opposes the cause responsible for its production.

**ii) Predict the direction of induced current in the conducting ring 1 when the current in the wire is steadily increasing**

According to Lenz's law, the induced current opposes the cause of increasing of magnetic flux. Direction of the current is clockwise in ring 1





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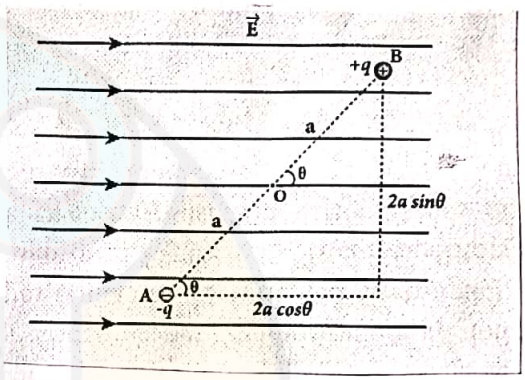
**26. Derive an expression for electrostatic potential energy of the dipole in a uniform electric field**

Consider a dipole placed in the uniform electric field as shown in the Figure

A dipole experiences a torque when kept in an uniform electric field $\vec{E}$.. This torque rotates the dipole to align it with the direction of the electric field. To rotate the dipole (at constant angular velocity) from its initial angle $\theta'$ to another angle $\theta$ against the torque exerted by the electric field, an equal and opposite external torque must be applied on the dipole.

The work done by the external torque to rotate the dipole from angle to $\theta$ at constant angular velocity is

$$W = \int_{\theta'}^{\theta} \tau_{ext}\, d\theta \ldots\ldots\ldots\ldots\ldots(1)$$




Since $\vec{\tau}_{ext}$ is equal and opposite to $\vec{\tau}_E = \vec{p} \times \vec{E}$ ,we have

$$|\vec{\tau}_{ext}| = |\vec{\tau}_E| = |\vec{p} \times \vec{E}| \ldots\ldots\ldots\ldots(2)$$

Substituting equation ( 2 ) in equation (1), we get

$$W = \int_{\theta'}^{\theta} pE \sin\theta\, d\theta$$

$$W = pE(-\cos\theta + \cos\theta')$$

$$W = pE(\cos\theta' - \cos\theta)$$

This work done is equal to the potential energy difference between the angular positions $\theta$ and $\theta'$.

$$U(\theta) - U(\theta') = \Delta U = -pE\cos\theta + pE\cos\theta'$$

If the initial angle $\theta'$ is 90° and is taken as reference point,

Then $U(\theta') = pE\cos 90° = 0$

The potential energy stored in the system of dipole kept in the uniform electric field is given by $U = -pE\cos\theta = -\vec{p}\cdot\vec{E}$

In addition to p and E, the potential energy also depends on the orientation $\theta$ of the electric dipole with respect to the external electric field.

The potential energy is maximum when the dipole is aligned anti-parallel ($\theta = \pi$) to the external electric field $U = pE$ $(\cos\pi = -1)$

and minimum when the dipole is aligned parallel ($\theta = 0$) to the external electric field.

$$U = -pE \quad (\cos 0° = 1)$$

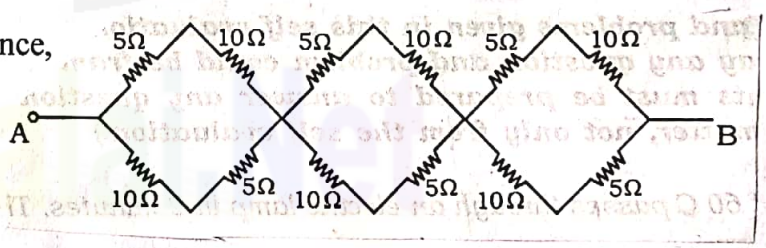
**27. Determine the effective resistance of the given circuit between points A and B.**

Resistors in series, the effective resistance,

$$R_s = R_1 + R_2$$

Resistors in parallel, the effective resistance,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$




Here $R_1$ and $R_2$ are in series

$$R_{s_1} = R_1 + R_2 = 5 + 10 = 15\Omega$$

Here $R_3$ and $R_4$ are in series,

$$R_{s_2} = R_3 + R_4 = 10 + 5 = 15\Omega$$

$R_{s_1}$ and $R_{s_2}$ are in parallel, $\frac{1}{R_p} = \frac{1}{R_{s_1}} + \frac{1}{R_{s_2}} = \frac{1}{15} + \frac{1}{15} = \frac{2}{15}$

$$R_p = \frac{15}{2} = 7.5\ \Omega$$

Total resistance of such three combination $= 3 \times 7.5 = 22.5\ \Omega$


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**28. i) Define power of the lens**

The power of a lens P is defined as the reciprocal of its focal length. $P = \frac{1}{f}$

**ii) Is it possible for two lenses in contact to produce zero power? Explain.**

The power of the combination of two lenses is given by the sum of the individual power of the lens.

If one of the lens is converging lens with focal length of power $P_1 = P$ and other diverging lens of focal length with negative sign of power $P_2 = -P$,

combination of these two lenses will be zero. $P_{Total} = P_1 + P_2 = 0$

**29. Can we completely replace humans with robots? Give three reasons.**

No, we can not completely replace humans with robots because

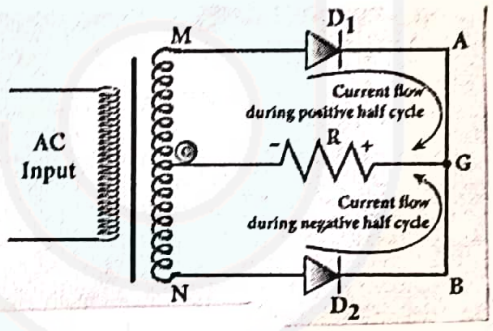
(i) Robats have no sense of emotions or conscience

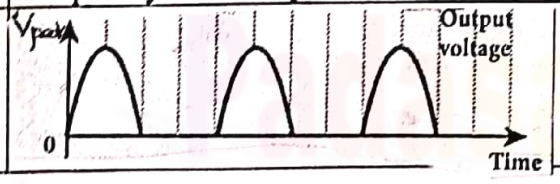
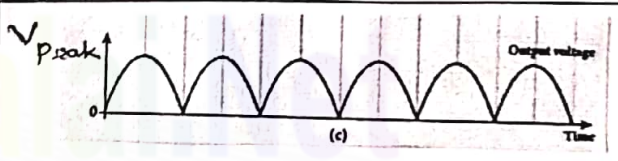
(ii) They lack empathy and hence create an emotionless workplace

(iii) Unemployment problem will increase.

(iv) Humans cannot be replaced by robots in decision making

**30. Draw the circuit diagram for full wave rectifier. Compare the output wave forms half-wave and full-wave rectifier.**




| S No | Output waveform of Half wave rectifier | Output waveform of Full wave rectifier |
|---|---|---|
| 1 | It has output only during positive cycle of the input | It has output both positive as well as negative half cycle of the input |
| 2 | Output frequency is equal to the frequency of the input | Output frequency is equal to twice of the input |
| 3 |   |   |

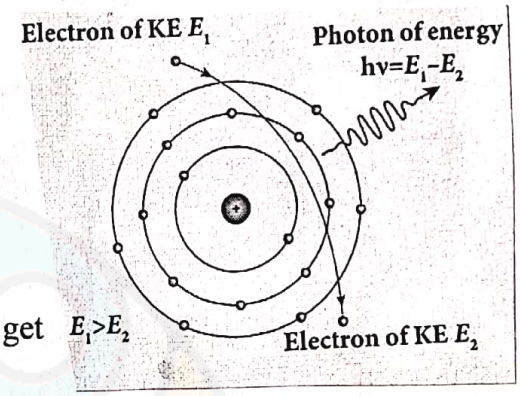
**31. Derive the Duane – Hunt formula for continuous X – rays. Can this be explained by classical electromagnetic theory?**

When a fast moving electron penetrates and approaches a target nucleus, the interaction between the electron and the nucleus either accelerates or decelerates it which results in a change of path of the electron. The radiation produced from such decelerating electron is called **Bremsstrahlung or braking radiation**

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The energy of the photon emitted is equal to the loss of kinetic energy of the electron. Since an electron may lose part or all of its energy to the photon, the photons are emitted with all possible energies (or frequencies). The continuous x-ray spectrum is due to such radiations.

When an electron gives up all its energy, then the photon is emitted with highest frequency $\nu_0$ (or lowest wavelength $\lambda_0$).




The initial kinetic energy of an electron = eV

where V is the accelerating voltage.

Therefore, we have

$$h\nu_o = eV \text{ (or) } \frac{hC}{\lambda_o} = eV$$

$$\lambda_o = \frac{hC}{eV}$$

where $\lambda_0$ is the cut-off wavelength.

Substituting the known values in the above equation, we get

$$\lambda_o = \frac{12400}{V} \text{ Å}$$

The relation is known as the **Duane – Hunt formula.**

This can be explained on the basis of **photon theory of radiation** not by classical electromagnetic theory

**32. Explain the spectral series of hydrogen which are in infra-red region of the electromagnetic waves**

The wave number or wavelength of spectral derived from Bohr atom model

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{n^2} - \frac{1}{m^2}\right)$$

| S.No | Spectral series | n | m | Wave Number | Region |
|---|---|---|---|---|---|
| 1 | Paschen series | 3 | 4,5… | $\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{m^2}\right)$ | Infra-red region (near IR) |
| 2 | Brackett series | 4 | 5,6….. | $\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{m^2}\right)$ | Infra-red region (middle IR) |
| 3 | Pfund series | 5 | 6,7… | $\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{m^2}\right)$ | Infra-red region (far IR) |

**33. Calculate the magnetic field at the centre of a square loop which carries a current of 1.5A, length of each loop is 50 cm.**

$$B = \frac{\mu_o I}{4\pi a}[\cos\phi_1 - \cos\phi_2]$$

$$= \frac{4\pi\times10^{-7}\times1.5}{4\pi\times0.25}[\cos 45° - \cos(180° - 45°)]$$

$$= \frac{10^{-7}\times1.5}{0.25}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$$

$$= 6\times10^{-7}\times\frac{2}{\sqrt{2}}$$

$$= 6\times10^{-7}\times\sqrt{2} = 8.484\times10^{-7}\text{Tesla}$$

For four sides $B' = 4\times8.484\times10^{-7} = 3.3936\times10^{-6}$ Tesla $\approx 3.4\times10^{-6}$ Tesla


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## PART - IV

**Answer all the questions** $5 \times 5 = 25$

**34.a) Derive the expression for radius of Snell's window.**

when light entering the water from outside is seen from inside the water, the view is restricted to a particular angle equal to the critical angle $i_c$. The restricted illuminated circular area is called *Snell's window*

The Snell's law in product form for the refraction happening at the point *B* on the boundary between the two media is,

$$n_1 \sin i_c = n_2 \sin 90^0 \quad \text{----------(1)}$$

$$n_1 \sin i_c = n_2 \qquad \because \sin 90^0 = 1$$

$$\text{Sin}i_c = \frac{n_2}{n_1} \quad \text{----------(2)}$$

From the right angle triangle ΔABC,

$$\text{Sin}i_c = \frac{CB}{AB} = \frac{R}{\sqrt{d^2+R^2}} \quad \text{----------(3)}$$

Equating the above two equation(2) and equation (3), $\frac{R}{\sqrt{d^2+R^2}} = \frac{n_2}{n_1}$

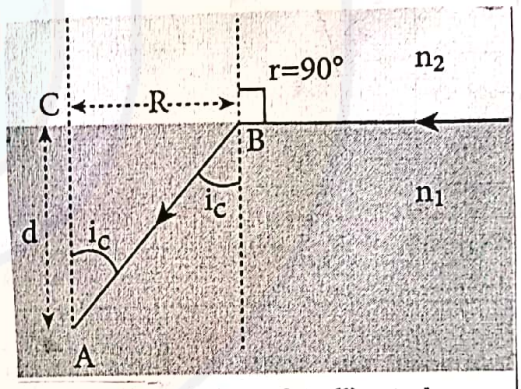
Squaring on both sides, $\frac{R^2}{R^2+d^2} = \left(\frac{n_2}{n_1}\right)^2$

Taking reciprocal, $\frac{R^2+d^2}{R^2} = \left(\frac{n_1}{n_2}\right)^2$

On further simplifying, $1 + \frac{d^2}{R^2} = \left(\frac{n_1}{n_2}\right)^2$

$$\frac{d^2}{R^2} = \left(\frac{n_1}{n_2}\right)^2 - 1$$

$$\frac{d^2}{R^2} = \frac{n_1^2}{n_2^2} - 1 = \frac{n_1^2-n_2^2}{n_2^2}$$



Radius of Snell's window

Again taking reciprocal and rearranging, $\frac{R^2}{d^2} = = \frac{n_2^2}{n_1^2-n_2^2}$

$$R^2 = d^2\left(\frac{n_2^2}{n_1^2-n_2^2}\right)$$

The radius of illumination is, $R = d\sqrt{\frac{n_2^2}{n_1^2-n_2^2}}$

If the rarer medium outside is air, then, $n_2 = 1$, and we can take $n_1 = n$

$$R = d\left(\sqrt{\frac{1}{n^2-1}}\right) \text{ (or) } R = \frac{d}{\sqrt{n^2-1}}$$

**(OR)**

**b. i) Obtain the expression for capacitance for a parallel plate capacitor**

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Consider a capacitor with two parallel plates each of cross-sectional area A and separated by a distance d as shown in the Figure

The electric field between two infinite parallel plates is uniform and is given by

$E = \frac{\sigma}{\varepsilon_o}$ where σ is the surface charge density on the plates .( $\sigma = \frac{q}{A}$)

If the separation distance d is very much smaller than the size of the plate ($d^2 << A$),then the above result is used even for finite–sized parallel plate capacitor.

The electric field between the plates is

$E = \frac{Q}{A\varepsilon_0}$ .....(1)

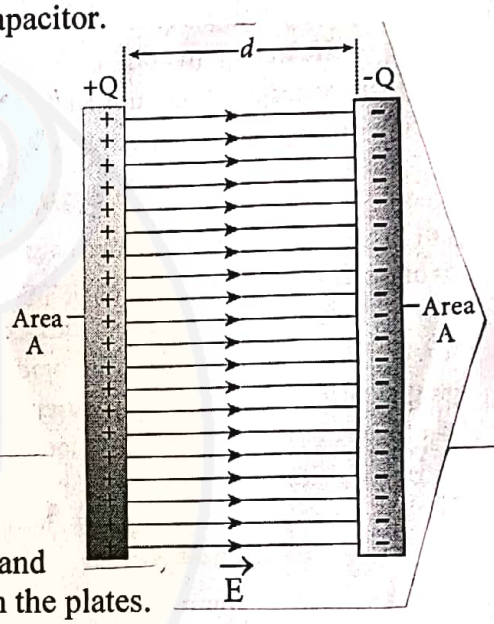
Since the electric field is uniform, the electric potential between the plates having separation d is given by

$V = Ed = \frac{Qd}{A\varepsilon_0}$ ..... (2)

Therefore the capacitance of the capacitor is given by

$C = \frac{q}{V} = \frac{Q}{\left(\frac{Qd}{A\varepsilon_0}\right)}$

$C = \frac{\varepsilon_0 A}{d}$.....(3)




From the equation (3), it is evident that capacitance is

(i) directly proportional to the area of cross section and

(ii) is inversely proportional to the distance between the plates.

**ii) Two capacitors of unknown capacitances are connected in series and parallel. If the net capacitances in the two combination are 6μF and 25μF respectively, find their capacitances.**

capacitors in series, the effective capacitance, $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$

$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6}$

$\frac{C_1+C_2}{C_1C_2} = \frac{1}{6}$ .....(1)

capacitors in parallel, the effective capatance, $C_p = C_1+C_2$

$C_p = C_1+C_2 = 25$.....(2)

Substitute (2) in (1) $\frac{25}{C_1C_2} = \frac{1}{6}$ ;

$C_1C_2 = 150$.....(3)

From(2) , we can write, $C_2 = 25 - C_1$ in (3)

$C_1(25 - C_1) = 150$

$C_1^2 - 25C_1 + 150 = 150$

$\therefore C_1 = 10$ (or) 15 μF

If $C_1 = 10$ μF and $C_2 = 15$ μF


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**35. a) Prove that the total energy remains constant during LC oscillations.**

During LC oscillations in LC circuits, the energy of the system oscillates between the electric field of the capacitor and the magnetic field of the inductor.

Although , these two forms of energy vary with time, the total energy remains constant LC oscillations take place in accordance with the law of conservation of energy.

$$\text{Total energy} = U = U_E + U_B = \frac{q^2}{2C} + \frac{1}{2} Li^2$$

Let us consider three different stages of LC oscillations and calculate the total energy of the system

case(i) When the charge $q = Q_m$, $i = 0$ ;

$$\text{Total energy} = U = \frac{Q_m^2}{2C} + 0 = \frac{Q_m^2}{2C}$$

The total energy is wholly electrical

case (ii) when $q = 0$, $i = I_m = Q_m\omega = \frac{Q_m}{\sqrt{LC}}$

$$\text{Total energy} = U = 0 + \frac{1}{2} LI_m^2 = \frac{L}{2} \times \left(\frac{Q_m^2}{LC}\right) = \frac{Q_m^2}{2C}$$

The total energy is wholly magnetic.

Case (iii) Charge $= q = Q_m \cos\omega t$

current $= i = -\frac{dq}{dt} = Q_m \omega \sin\omega t$

Negative sign in current indicates that the charge in the capacitor decreases with time

$$\text{Total energy} = U = \frac{q^2}{2C} + \frac{1}{2} Li^2$$

$$= \frac{Q_m^2 \cos^2\omega t}{2C} + \frac{L\omega^2 Q_m^2 \sin^2\omega t}{2}$$

$$= \left(\frac{Q_m^2 \cos^2\omega t}{2C} + \frac{LQ_m^2 \sin^2\omega t}{2LC}\right) \qquad \because \omega^2 = \frac{1}{LC}$$

$$= \frac{Q_m^2 \cos^2\omega t}{2C} + \frac{Q_m^2 \sin^2\omega t}{2C}$$

$$= \frac{Q_m^2}{2C} (\sin^2\omega t + \cos^2\omega t)$$

$$= \frac{Q_m^2}{2C}$$

From above three cases, it is clear that the total energy of the system remains constant

**(OR)**

**b) With proper explanations derive the Einstein's photoelectric equation. Draw the Frequency – $K_{max}$ graph and derive the equation for $K_{max}$ of the electron.**

When a photon of energy $hv$ is incident on a metal surface, it is completely absorbed by a single electron and the electron is ejected.



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In this process, (i) a part of the photon energy is used for the ejection of the electrons from the metal surface (photoelectric work function $\phi_0$ ) and
(ii) the remaining energy as the kinetic energy of the ejected electron.

From the law of conservation of energy, $h\nu = \phi_0 + \frac{1}{2} mv^2$----------(1)

where $m$ - mass of the electron and
$\upsilon$ - velocity.

At threshold frequency $\nu_0$ of incident radiation, the photo electrons are ejected with almost zero kinetic energy

Then the equation (1) becomes $h\nu_0 = \phi_0$

By rewriting the equation ( 1), we get $h\nu = h\nu_0 + \frac{1}{2} mv^2$ ----------(2)

The equation (2) is known as **Einstein's photoelectric equation.**

If the electron does not lose energy by internal collisions, then it is emitted with maximum kinetic energy $K_{max}$.

Then

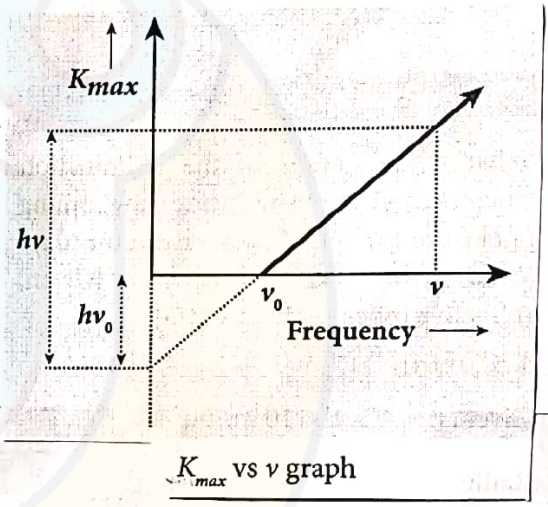
$$K_{max} = \frac{1}{2} mv_{max}^2$$

where $v_{max}^2$ is the maximum velocity of the electron ejected.

The equation (1) is rearranged as follows:

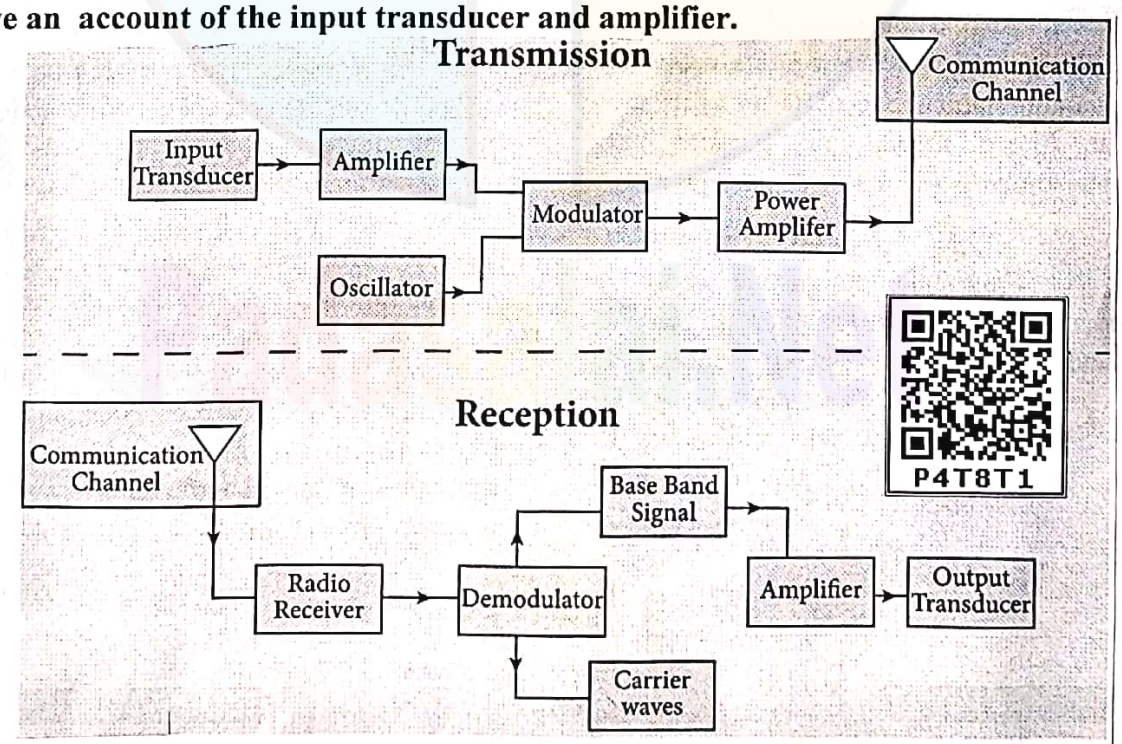
$$K_{max} = h\nu - \phi_0$$

A graph between maximum kinetic energy $K_{max}$ of the photoelectron and frequency $\nu$ of the incident light is a straight line as shown in Figure




$K_{max}$ vs $\nu$ graph

**36) a) With the block diagram to explain the transmission and reception of voice signals , give an account of the input transducer and amplifier.**




Block diagram of transmission and reception of voice signals


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**Input transducer**

A transducer is a device that converts variations in a physical quantity (pressure, temperature, sound) into an equivalent electrical signal or vice versa. In communication system, the transducer converts the information which is in the form of sound, music, pictures or computer data into corresponding electrical signals.

**The electrical equivalent of the original information is called the baseband signal.**

The best example for the transducer is the microphone that converts sound energy into electrical energy.

**Amplifier**: The transducer output is very weak and is amplified by the amplifier.

**Power amplifier**: It increases the power level of the electrical signal in order to cover a large distance.

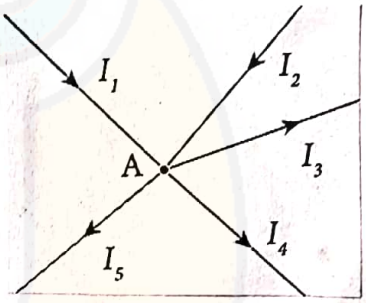
**(OR)**

**b) State Kirchhoff's firtst law and second law and explain the sign conventions to be followed while using the laws.**

**Kirchhoff's current rule**

It states that the algebraic sum of the currents at any junction of a circuit is zero.

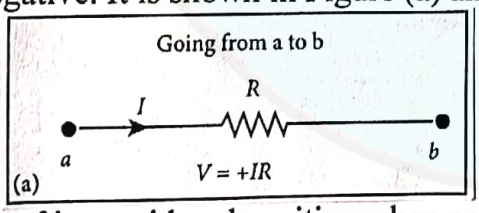
Current entering the junction is taken as positive and current leaving the junction is taken as negative.

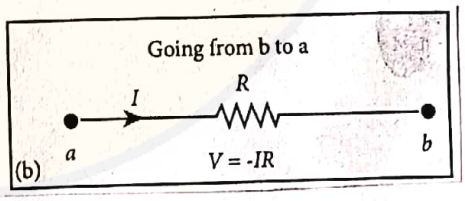



**Kirchhoff's voltage rule**

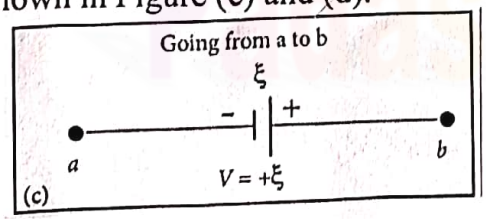
It states that in a closed circuit, the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit.

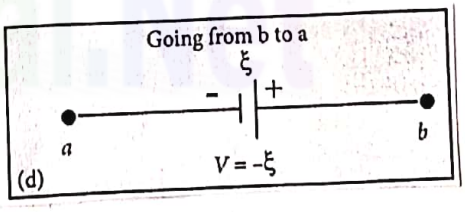
The product of current and resistance is taken as positive when the direction of the current is followed. Suppose if the direction of current is opposite to the direction of the loop, it is taken as negative. It is shown in Figure (a) and (b).




The emf is considered positive when proceeding from the negative to the positive terminal of the cell and negative when proceeding from the positive to the negative terminal of the cell . It is shown in Figure (c) and (d).



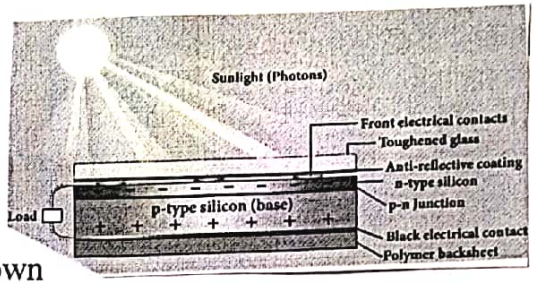




**37.a) What is solar cells? Draw the cross sectional view of solar cell and explain its construction and working. Write a note on solar arrays.**

A solar cell, also known as photovoltaic cell, converts light energy directly into electricity or electric potential difference by **photovoltaic effect.**


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It is basically a p-n junction which generates emf when solar radiation falls on the p-n junction.




**Construction**

A solar cell is of two types: p-type and n-type.
Both types use a combination of p-type and n-type Silicon which together forms the p-n junction of the solar cell.
The difference is that p-type solar cells use p-type Silicon as the base with an ultra-thin layer of n-type Silicon as shown in Figure, while n-type solar cell uses the opposite combination.
The other side of the p-Silicon is coated with metal which forms the back electrical contact.
On top of the n-type Silicon, metal grid is deposited which acts as the front electrical contact.
The top of the solar cell is coated with anti-reflection coating and toughened glass.

**Working**

In a solar cell, electron–hole pairs are generated due to the absorption of light near the junction.
Then the charge carriers are separated due to the electric field of the depletion region.
Electrons move towards n–type Silicon and holes move towards p-type Silicon layer.
The electrons reaching the n-side are collected by the front contact and holes reaching p-side are collected by the back electrical contact. Thus a potential difference is developed across solar cell.
When an external load is connected to the solar cell, photocurrent flows through the load.
Many solar cells are connected together either in series or in parallel combination to form **solar panel or module.**
Many solar panels are connected with each other to form **solar arrays.** For high power applications, solar panels and solar arrays are used.

**(OR)**

**b) What is meant by displacement current? Write the Maxwell equations in integral form and explain them**

**Displacement current** can be defined as the current which comes into play in the region in which the electric field and the electric flux are changing with time

Electrodynamics can be summarized into four basic equations, known as Maxwell's equations. These equations are analogous to Newton's equations in mechanics. Maxwell's equations completely explain the behaviour of charges, currents and properties of electric and magnetic fields. These equations can be written in integral .

1. First equation is nothing but the Gauss's law. It relates the net electric flux to net electric charge enclosed in a surface. Mathematically, it is expressed as

$$\oint \vec{E} \cdot \overrightarrow{dA} = \frac{Q_{enclosed}}{\varepsilon_0} \quad \text{( Gauss's law )}$$

Where $\vec{E}$ is the electric field and $Q_{enclosed}$ is the charge enclosed.

This equation is true for both discrete or continuous distribution of charges.
It also indicates that the electric field lines start from positive charge and terminate at negative charge.
This implies that the electric field lines do not form a continuous closed path.
In other words, it means that isolated positive charge or negative charge can exist.


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2. Second equation has no name. But this law is similar to Gauss's law in electrostatics. So this law can also be called as Gauss's law in magnetism. The surface integral of magnetic field over a closed surface is zero. Mathematically,

$$\oint \vec{B} \cdot \overrightarrow{dA} = 0 \text{ ( No name)}$$

Where $\vec{B}$ is the magnetic field.

This equation implies that the magnetic lines of force form a continuous closed path.

In other words, it means that no isolated magnetic monopole exists.

3. Third equation is Faraday's law of electromagnetic induction. This law relates electric field with the changing magnetic flux which is mathematically written as

$$\oint \vec{E} \cdot \overrightarrow{dl} = -\frac{d}{dt} \Phi_B \text{ (Faraday's law)}$$

where $\vec{E}$ is the electric field.

This equation implies that the line integral of the electric field around any closed path is equal to the rate of change of magnetic flux through the closed path bounded by the surface. Our modern technological revolution is due to Faraday's laws of electromagnetic induction. The electrical energy supplied to our houses from electricity board by using Faraday's law of induction.

4. Fourth equation is modified Ampere's circuital law. This is also known as Ampere – Maxwell's law. This law relates the magnetic field around any closed path to the conduction current and displacement current through that path.

$$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 I_{enclosed} + \mu_0 \varepsilon_0 \frac{d}{dt} \int \vec{E} \cdot \overrightarrow{dA} \text{ (Ampere Maxwell's law)}$$

where $\vec{B}$ is the magnetic field.

This equation shows that both conduction and also displacement current produces magnetic field.

These four equations are known as Maxwell's equations in electrodynamics.

This equation ensures the existence of electromagnetic waves.

**38) i) Write a short note on carbon dating**

ω The interesting application of beta decay is radioactive dating or carbon dating.

Using this technique, the age of an ancient object can be calculated.

All living organisms absorb carbon dioxide ($CO_2$) from air to synthesize organic molecules.

In this absorbed $CO_2$, the major part is ${}^{12}_{6}C$ and very small fraction ($1.3 \times 10^{-12}$ )is radioactive ${}^{14}_{6}C$ whose half-life is 5730 years.

But when the organism dies, it stops absorbing $CO_2$. Since ${}^{14}_{6}C$ to decay, the ratio of ${}^{14}_{6}C$ to ${}^{12}_{6}C$ in a dead organism or specimen decreases over the years. Suppose the ratio of ${}^{14}_{6}C$ to ${}^{12}_{6}C$ in the ancient tree pieces excavated is known, then the age of the tree pieces can be calculated. This is known as carbon dating.

**ii) A radio actie sample has 2.6 μg of pure ${}^{13}_{7}N$ which has a half life of 10 minutes. How many nuclei are present initially? Also find its initial activity?**


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$^{13}_{7}N$ atom consists of $6.023 \times 10^{23}$ atoms in 13g

Number of atoms in 1 gram $= \frac{6.023\times10^{23}}{13}$

Number of atoms in $2.6 \times 10^{-6}$ gram $= N_o = \frac{6.023\times10^{23}}{13} \times 2.6 \times 10^{-6} = 12.04 \times 10^{16}$ atoms

Activity $= R_o = \lambda N_o$

$= \frac{0.6931}{T_{\frac{1}{2}}} \times N_o$ $\quad \because T_{\frac{1}{2}} = \frac{0.6931}{\lambda}$

$= \frac{0.6931}{10\times60} \times 12.04 \times 10^{16}$

$= 1.391 \times 10^{14}$ Bq

$= \frac{1.391\times10^{14}}{3.7\times10^{10}} = 0.3759 \times 10^{4}$ $\quad \because$ 1 curie $= 3.7 \times 10^{10}$ Bq

$= 3.759 \times 10^{3}$ Ci

**(OR)**

**b) What is cyclotron?. Explain in detail the principle, construction and working and obtain the equation for the time period of the particle(Draw the cyclotron diagram with parts)**

Cyclotron is a device used to accelerate the charged particles to gain large kinetic energy. It is also called as high energy accelerator. It was invented by Lawrence and Livingston in 1934.

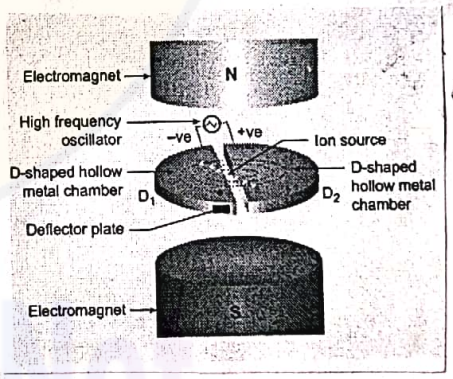
**Principle:** When a charged particle moves normal to the magnetic field, it experiences magnetic Lorentz force

**construction**

The schematic diagram of a cyclotron is shown in Figure

The particles are allowed to move in between two semi-circular metal containers called Dees (hollow D - shaped objects).

Dees are enclosed in an evacuated chamber and it is kept in a region with uniform magnetic field controlled by an electromagnet. The direction of magnetic field is normal to the plane of the Dees.




The two Dees are kept separated with a gap and the source S is placed at the center in the gap between the Dees.

Dees are connected to high frequency alternating potential difference.


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**Working**

Let us assume that the ion ejected from source S is positively charged. As soon as ion is ejected, it is accelerated towards a Dee (say, Dee – 1) which has negative potential at that time. Since the magnetic field is normal to the plane of the Dees, the ion undergoes circular path.

After one semi-circular path in Dee-1, the ion reaches the gap between Dees. At this time, the polarities of the Dees are reversed so that the ion is now accelerated towards Dee-2 with a greater velocity.

For this circular motion, the centripetal force of the charged particle q is provided by Lorentz force.

$$\frac{mv^2}{r} = qvB$$

$$\Longrightarrow r = \frac{m}{qB} v$$

$$\Longrightarrow r \propto v \quad \text{----------(1)}$$

From the equation (1), the increase in velocity increases the radius of circular path. This process continues and hence the particle undergoes spiral path of increasing radius.

Once it reaches near the edge, it is taken out with the help of deflector plate and allowed to hit the target T.

Very important condition in cyclotron operation is the resonance condition. It happens when the frequency $f$ at which the positive ion circulates in the magnetic field must be equal to the constant frequency of the electrical oscillator $f_{osc}$.

$$f_{osc} = \frac{qB}{2\pi m}$$

The time period of oscillation is $T = \frac{2\pi m}{qB}$

T. JOHN PETER,
St. Joseph's HSS, Cuddalore-1
9566551159

Padasalai.Net



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